MIL-B-36939
16 February 1973

MILITARY SPECIFICATION

BLOOD COLLECTING-DISPENSING BAG

AND DONOR SET WITH 63 cc ANTICOAGULANT, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a plastic blood collecting and dispensing bag with anticoagulant solution and integral donor set, suitable for collection, storage, processing, and administration of human blood. Potency period is 48 months.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-N-196	Needles, Hypodermic, for Luer Syringes.
PPP-B-636	Box, Fiberboard.
PPP-C-96	Can, Metal, 28 Gage and Lighter.

Military

MIL-B-36646	Blood Recipient Set, with Metering Chamber, Without Needle, Disposable.

STANDARDS

Federal

FED-STD-142	Parenteral Preparations.

FSC 6515

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (U.S.P.)

(Application for copies shall be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042).

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Federal Food, Drug, and Cosmetic Act and General Regulations for its Enforcement.

(Application for copies should be addressed to the Food and Drug Administration, U.S. Department of Health, Education, and Welfare, Washington, D.C. 20204).

NATIONAL INSTITUTES OF HEALTH

Public Health Service Regulations-Part 73* Pertaining to Federal Control of Sale, Barter, or Exchange of Biological Products.

*(Application for copies should be addressed to the Laboratory of Biologics Control National Instututes of Health, Bethesda, Maryland 20034).

3. REQUIREMENTS

3.1 Material. All material used in the fabrication of the blood collecting and dispensing set shall be of the best quality and suitable in all respects for the intended purpose. The set shall be of such nature or so treated to prevent growth of molds or spore-bearing organisms. All plastic material coming in contact with blood and components shall meet the requirements of the United States Pharmacopeia (U.S.P.) for Class II plastic, when tested as specified in 4.4.1, and shall contain not more than 5 parts per million (p.p.m.) total of barium, cadmium, and lead, when tested as specified in 4.4.2.

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3.1.1 Plastic bag material. The plastic of the bag shall be transparent, flexible, sterilizable, nontoxic, and relatively nonfrangible. The plastic shall be compatible with the liquids to be contained as specified herein. It shall have no appreciable deleterious effect upon the viability of red blood cells stored in the bag as required by the Federal Food, Drug, and Cosmetic Act. The plastic shall not become tacky following sterilization or during storage for a minimum period of four years.

3.1.1.1 Thickness. The sheet material of individual bags shall have a thickness of from 0.008 to 0.032 inch (in).

3.1.1.2 Transparency. The transparency of the bag shall be such that visual inspection of the contents before and during collection, during processing, storage and administration shall be possible under normal illumination.

3.1.1.3 Flexibility. The flexibility of the plastic material shall be sufficient to permit fabrication of a collapsible bag, which shall offer a minimum resistance to filling and emptying of the bag without air exchange. This flexibility shall be maintained at temperatures of 0° Centrigrade (C.). to 50° C.

3.1.1.4 Elasticity. The elasticity of the bag material shall be such that the finished bag, filled with 450 milliliters (ml.) of blood and 63 ml. of anticoagulant solution, shall be capable of withstanding centrifugation at 2500 times gravity in a conventional centrifuge cup, and of withstanding indentation without rupture by wire shipping racks.

3.1.1.5 Thermal stability. The plastic material shall be such that the finished bag, containing anticoagulant solution, shall be capable of withstanding storage without deleterious effect at ambient temperatures of from minus 50°C. to + 50°C. The material of the bag shall be such as to permit sterilization by autoclaving prior to use.

3.1.1.6 Vapor transmissivity. The vapor transmissivity of the bag material shall be such that the finished, sterile bag shall not lose more than 25 ml. of water, when tested as specified in 4.4.3.

3.1.1.7 Leaching. Each lot of plastic sheeting shall have a weight loss of not more than 0.35 percent, when tested as specified in 4.4.4.

3.1.2 Anticoagulant solution. The anticoagulant solution shall conform to Anticoagulant Citrate Phosphate Dextrose Solution in accordance with the U.S.P. and the Additional Standards: Whole Blood (human) of the Public Health Service Regulations-Part 73. The anticoagulant solution shall be in accordance with the requirements of Fed. Std. No. 142a for type I, class 1.

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3.1.3 Tubing for donor set. The tubing for the donor set shall be fabricated from nontoxic, transparent, pliable plastic. The tubing shall be sufficiently rigid to prevent collapse during normal use and of such flexibility that the bag can be moved during the phlebotomy without unduly disturbing the needle. The vapor transmissivity of the donor tube attached to the bag shall be as specified in 3.1.1.6. Testing of the fabricated unit as in 3.1.1.6 shall constitute compliance with this requirement. The plastic tubing shall not become tacky following sterilization or during storage for a minimum period of four years.

3.1.4.1 Cannula. The needle cannula shall be 15 or 16 gage, thin wall, 1-1/2 inch long tubing fabricated of AISI 304 corrosion-resisting steel. The cannula shall pass the corrosion-resistance, stiffness, elasticity, and resistance to breakage tests specified in GG-N-196 for the 17-gage needle size. Elasticity and resistance to breakage tests shall be performed on the cannula with the hub provided on the finished sample set. The needle may be treated with silicone, and in addition not more than 1.0 mg of total residue per needle shall be present when determined by appropriate, accurate, and reproducible method.

3.1.4.2 Hub. The needle hub shall be fabricated of material suitable in all respects for the intended purpose and compatible with the needle cannula, the plastic tubing, and the anticoagulant solution.

3.2 Design of set. The design of the blood collecting and dispensing set shall be such as to provide for collection, storage, and processing of blood, and permit removal of plasma and administration of whole blood or centrifuged or resuspended blood cells, with minimal hazard of bacterial contamination. The set shall consist of a plastic bag with anticoagulant solution, two outlet ports, and an inlet port with integrally attached donor set, all sterilized as one unit. The design of the set shall be such as to permit the collection of pilot samples of unmistakable identity for grouping and typing, cross-matching, and culturing. These pilot samples shall be obtained without the necessity of entering the bag proper. A side pouch shall not be incorporated in the set design (i.e., attached to tubing).

3.2.1 Plastic bag. The inside overall dimensions of the bag, after sterilization, shall be 7 $\pm$ 3/8 in. long by 4-3/4 $\pm$ 1/8 in. wide, when measured flat. The shape of the bag shall permit centrifugation in a standard trunnion cup. The bag shall be provided with an inlet port, to which a donor set is permanently attached and sealed, and two outlet ports, separate from the inlet port. Each outlet port shall contain a diaphragm which can be readily punctured in order to couple and administer blood and its components through a standard blood recipient set in accordance with MIL-B-36646. The inlet port shall be provided with a device which shall prevent influx of air when the needle protector sleeve is removed. This device shall allow free flow of the blood being collected and shall not impede the flow into the blood recipient set in accordance with MIL-B-36646 by blocking the piercing connectors. The arrangement of the donor tube and outlet ports, as they are attached to the bag shall be such that,

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3.2.1 Plastic bag (con't)

when the bag is in its upright position, these connections shall be at the top of the bag. Their arrangement shall provide maximum efficiency for component separation and for whole blood and blood component administration. Two suitable holders shall be provided on each side of the bag for holding blood collection tubes in the upright position. The unit shall not be weakened by the presence of the holders. The side seals shall not be damaged by the insertion of the tubes.

3.2.1.1 Ports and closure. The outlet ports shall consist of externally protruding tubes hermetically sealed to the bag. The entire exterior surface of the protruding tubes shall be sterile and have its sterility maintained by a tamperproof protective sleeve or rubber plug. The outlet ports shall be of such design as to hold and make a leakproof fit with a standard blood recipient set in accordance with MIL-B-36646. In addition, the outlet ports shall make a leakproof fit with the blood transfer bag FSN 6515-890-1866 and with any recipient set whose piercing connector has a minimum length of 1-1/4 in. and a mean external diameter of not less than 0.208 in. The outlet ports shall be so constructed as to protect the walls of the bag against piercing by the recipient set connector. The ports shall be so attached to the bag and so reinforced as to withstand any pulling or twisting encountered during normal usage. All closures of the bag shall be hermetic. These closures shall be so designed as to permit blood collection, storage, transportation, and administration without contamination.

3.2.1.2 Bag suspension. Means of suspending the bag shall be incorporated in the bag design. The means of suspending the bag during the administration of blood shall be a horizontal slot, 1-1/4 $\pm$ 1/8 in. long, in the seam of the plastic at the bottom of the bag. The bag may have circular holes, 1/4 $\pm$ 1/64 in. in diameter, located 1/4 $\pm$ 1/64 in. from each corner in the seam at the bottom of the bag for corner hanging position. The bag shall have 2 circular holes, 1/8 $\pm$ 1/64 in. in diameter and not more than 2 in. apart, centrally located in the seam at the top of the bag to permit suspension in an upright position during storage or collection of blood. As an alternate to the circular holes in the top of the bag, there may be 2 loops formed from plastic strips, each approximately 1/4 in. wide in the top of the bag to permit suspension in an upright position. These loops shall be centrally located, one on each side of the central (inlet or outlet) member. When tested as specified in 4.4.5, the bag shall not rupture and the suspension slot and loops shall not sever from the bag.

3.2.1.3 Bag strength. The finished, sterile plastic bag shall be free of leaks and shall not develop leaks when filled and subjected to a pressure of 500 millimeters (mm.) of mercury at 25°C., when tested as specified in 4.4.6.

3.2.1.4 Bag contents. The finished bag shall contain 63 ml. of anticoagulant solution specified in 3.1.2. The bag shall contain not more than 5.0 cubic centimeters (cc.) of air corrected to a pressure of 760 mm. of mercury and at a temperature of 0°C. The anticoagulant solution shall be distributed in the bag proper and the donor tubing. The 5 cc. of air shall include the air in the donor tubing.

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3.2.1.5 Filling and infusion. The design of the bag shall be such that it shall be capable of being filled with 450 ml. of blood without the trapping of air (in addition to the 5 cc. permissible as specified in 3.2.1.4). When tested as specified in 4.4.7, the bag shall fill by gravity in less than 10 minutes at room temperature with the donor needle in the same hyrostatic plane as the top of the bag. When tested as specified in 4.4.8, the full bag shall empty completely in 2 minutes or less.

3.2.2 Donor set. The donor set shall consist of a length of plastic tubing and a donor needle with sleeve.

3.2.2.1 Donor tubing. The donor tubing shall consist of plastic tubing not less than 27 in. in length with an external diameter of not more than 3/16 in. The tubing shall be capable of withstanding an external pressure of 75 mm. of mercury without collapsing and shall be of such flexibility as to permit the tubing to be easily knotted. One end of the tubing shall be permanently sealed to the inlet port of the bag and the other end shall be fitted with a donor needle. The attached tubing shall not become detached from the bag or show any evidence of separation from the bag when tested as specified in 4.4.9.1. The tubing shall not become detached from the needle hub, nor shall the cannula separate from the hub, when tested as specified in 4.4.9.2. The donor tubing of each bag shall bear identical numbers and marks ("X's" or "*'s") alternately and indelibly imprinted along the tubing for the purpose of dividing the donor tubing into segments for pilot samples. The center-to-center distance of these marks shall be not less than 2-1/2 in. nor more than 3 in. The donor tubing of each unit shall bear a different number. The final number shall be immediately adjacent to the bag so that, at the time the donor tubing is severed, the identifying number remains with the bag.

3.2.2.2 Donor needle. The donor needle shall be an intravenous needle with a 15-gage or 16 gage, thin-wall, 1-1/2 in. cannula. The hub of the needle shall be of such size that it will form a tight, leakproof connection with the plastic donor tubing and of such design as to indicate, by finger feel or by marking, the plane of orientation of the bevel of the point. When tested as specified in 4.4.9.3, the cannula shall not move in, move out, or separate from the hub. The point of the needle shall have a sharp, smoothly ground bevel not less than 13/64 in. nor greater than 17/64 in. in outside length. The interior lumen of the connection between the needle cannula and the donor tube shall present a smoothly expanding surface to minimize the formation of eddy currents to the extent of preventing platelet aggregation and clotting. All interior and exterior surfaces of the needle cannula which come into contact with blood shall be hemo repellent to the extent of preventing platelet aggregation and clotting. The cannula of the needle shall be housed in a sheath or sleeve fabricated of plastic or rubber. The sheath or sleeve shall be so designed, and shall be so fitted to the cannula, that the sterility of the cannula and the assembled donor set and bag shall be maintained, and leakage of the anticoagulant solution out of the tip of the needle shall be prevented. The sheath or sleeve shall be of such size that it can be easily removed, but not fall off accidentally. The sheath or sleeve shall make an interference sealed fit with an internal surface of the needle hub in such a manner that only sterile surfaces and edges of the sheath or sleeve can contact the point of the needle cannula during removal. The needle shall be free from foreign matter upon normal visual examination after equipment has been sterilized.

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3.3 Compatibility. The materials of which the blood collecting and dispensing sets are manufactured shall not react with, or be acted upon by, the liquids with which they will be in contact during normal use and operation, nor shall there be any materials incorporated in the unit which will cause any harmful or deleterious effect when the unit is used for the purpose intended.

3.4 Sterility, safety, and freedom from pyrogens. The anticoagulant solution, bag interior, fluid path, entire needle surface, shall be sterile when tested as specified in 4.4.10 and pyrogen free when tested as specified in 4.4.11. The blood collecting and dispensing sets shall be safe (non-toxic) when tested for safety as specified in 4.4.12.

3.5 In vivo survival of red blood cells. The absence of demonstrable detrimental effect on the red cells shall be established by in vivo evidence. The viability of red cells of whole blood collected in the equipment shall be such that, after 21 days of storage at 5° ± 1°C., not less than 70 percent of the red cells are found to be viable, by in vivo testing as specified in 4.4.13. No change in formulation or fabrication of the plastic shall be permitted without submission of evidence that it does not decrease in vivo survival of red cells to less than the level specified above.

3.6 Certificates. For destination inspection only, certificates of sterility, safety, freedom from pyrogens, in vivo, red blood cell survival, and biological tests, as specified below, shall be submitted by the contractor directly to Technical Operations Division, Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa., 19101, Attention: Quality Control Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via the government inspector.

3.6.1 Certificate of sterility. A certificate of sterility, listing all sterility lot or control numbers, shall state that samples from each sterility lot or control have been tested for sterility, as required in 3.4, and were found to be sterile. The certificate shall reference the method used in determining sterility.

3.6.2 Certificates of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety, as specified in 3.4, and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.6.3 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens, as specified in 3.4, and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

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3.6.4 Certificate of in vivo red blood cell survival. A certificate of in vivo red blood cell survival, listing all plastic lot or batch numbers, shall state that samples from each lot or batch have been tested for in vivo red blood cell survival, as specified in 3.5, and were found to comply with such requirements. The certificate shall reference the method used in determining in vivo red blood cell survival.

3.6.5 Certificate of biological tests. A certificate listing all lot or control numbers shall state that samples from each lot or control have been tested and have passed the biological tests specified in the U.S.P. for class II plastic. This certificate shall include the specific tests performed and the test results.

3.7 Labeling. Each bag shall bear a label as shown in Figure 1. This label shall be securely attached to the bag and shall not be capable of removal without mutilation. When tested as specified in 4.4.14, the label shall not separate from the bag. The label shall not be secured by any method that would puncture or mutilate the bag. One side of the bag must remain visible and free of markings. The label shall be of such material and so applied to the bag that the label can be written upon legibly and that other labels can be adhered thereto and remain adhered throughout the usage time of the bag. Labeling shall conform with the requirements of the Federal Food, Drug and Cosmetic Act. Shall be labeled in the metric system and the U.S. Customary Systems of Weights and Measures. When it is current industry practice to label in the metric system, the item is not required to be labeled in the U.S. Customary system of Weights and Measures.

3.8 Instructions for use. The unit package shall be permanently and legibly marked with complete instructions for use in both collection and administration of blood as they pertain to the particular set.

3.9 Public Health Service Regulations. The equipment shall be in strict accordance with Additional Requirements: Whole Blood (Human) of the Public Health Regulations - Part 73.

3.10 Compliance with the Federal Food, Drug, and Cosmetic Act. The items furnished under this specification must meet the requirements of the Federal Food, Drug, and Cosmetic Act. The contractor shall submit evidence to the purchasing agency prior to award of contract that the items he proposes to furnish under this specification meet the requirements of the Federal Food, Drug and Cosmetic Act, including an approved New Drug Application on file with the Food and Drug Administration.

3.11 Manufacturer's record. The bag label specified in 3.7 shall be permanently and legibly marked with a code or lot number which can be verified against the manufacturer's records of all sterility, safety, freedom from pyrogens, red blood cell survival, and biological tests performed on the blood collecting and dispensing sets. The manufacturer's records shall be made available for inspection by any duly authorized government representative. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

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3.12 Workmanship. Workmanship shall be first class throughout. The blood collecting and dispensing bag (sets) shall be free from defects which detract from their appearance or may impair their serviceability.

3.13 Delivery. Not less than 44 months of the expiration dating period (potency period) shall remain at time of delivery to the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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FEDERAL STOCK NO. 6515-079-9483

BLOOD COLLECTING-DISPENSING BAG AND DONOR SET, with anticoagulant, 450 cc, Disposable

CPD WHOLE BLOOD
(HUMAN)

Contains 450 cc. of human blood plus 63 cc. of Anticoagulant Citrate Phosphate Dextrose Solution (CPD)

Donor Number

ABO Blood Group:

CAUTION

1. Store blood continuously within 2° range between 1° and 6°C. (34° and 43°F.). Keep within 1° and 10°C. (34° and 50°F.) during shipment
2. Cross match before using.
3. Invert unit several times before using to resuspend red cells.
4. Always administer through a filter.
5. Administer without warming.
6. Read label carefully to assure that Blood Group and Type are the same as those of the recipient. Check this with one other person.
7. Do not add other medication to the bag of blood prior to administration.
8. Read instructions on unit package before using.

RH Type:

Do Not Infuse After

Serology Non-Reactive to _____ Test

Mfr's name, address (if the mfr, is not the contractor, the contractor's name and address shall also appear); CPD formula; "Sterile, Pyrogen free", Fluid Path etc., and such other information as required by Federal Laws.

Collecting Activity:

Figure 1. Label for Blood Bag

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastic, the anticoagulant solution, or the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Lot.

4.2.1.1 Plastic. For purposes of this specification, a lot or batch shall consist of a quantity of plastic sheeting or tubing of uniform composition, produced under like conditions and shall represent one continuous calendering or extrusion run of plastic. In the case of sheeting, a continuous run may yield as much as 250,000 pounds.

4.2.1.2 Anticoagulant solution. For purposes of this specification, a lot or batch shall be as specified in FED-STD-142.

4.2.2 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table I. The unit of product for examination purposes shall be one blood collecting-dispensing bag and donor set.

Table I. Sampling for Examination

Defect category	Inspection level	AQL (percent defective)
Critical	II	0.65
Major	II	1.0
Minor	II	2.5

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4.2.3 For tests.

4.2.3.1 For sterility, pyrogen, and safety tests. Sampling for sterility, pyrogen, and safety tests shall be conducted as indicated in the United States Pharmacopeia (U.S.P.) for transfusion and infusion assemblies.

4.2.3.2 For in vivo test. Each manufacturer's lot of plastic sheeting shall be tested as specified in 4.4.13. If the material fails more than one test of the three (or four) tests required to be performed, the plastic batch represented shall be rejected.

4.2.3.3 For other tests. Sampling for all other tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Table II. Sampling Criteria for Tests

Component	Characteristic	Requirement	Test procedure	Unit of product	Insp. level	AQL (percent defective)
Set	Material-Class II	3.1	4.4.1	1-set	**	*
	Heavy metals	3.1	4.4.2	1-set	***	*
	Vapor transmissivity	3.1.3 and 3.1.1.6	4.4.3	1-set	S-1	*
	Leaching	3.1.1.7	4.4.4	****	S-2	*
	Dimensions	3.2.1, 3.2.1.2, and 3.2.2.1	- - -	1-set	S-4	1.5
	Bag suspension	3.2.1.2	4.4.5	1-set	S-3	1.5
	Bag strength	3.2.1.3	4.4.6	1-set	S-4	1.5
	Fluid fill	3.2.1.5	4.4.7	1-set	S-4	1.5
	Pressure infusion	3.2.1.5	4.4.8	1-set	S-4	1.5
	Security of attachment:					
	tubing to bag	3.2.2.1	4.4.9.1	1-set	S-3	1.5
	tubing to needle hub	3.2.2.1	4.4.9.2	1-set	S-3	1.5
	Label ashesion	3.7	4.4.14	1-set	S-3	2.5

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Table II. Sampling Criteria for Tests (Cont'd)

Component	Characteristic	Requirement	Test procedure	Unit of product	Insp. level	AQL (percent defective)
Donor needle	Corrosion resistance, stiffness, elasticity, and resistance to breakage of cannula	3.1.4.1	GG-N-196	1-needle	S-1	*
	Dimensions	3.2.2.2	- - - -	1-needle	S-4	1.5
	Security of assembly of cannula to hub	3.2.2.2	4.4.9.3	1-needle	S-3	1.5
	Foreign matter	3.2.2.2	- - - -	1-set	S-4	1.5

* Acceptance number shall be zero for all sample sizes.
** Each lot of product need not be biologically tested provided evidence is furnished that samples of the formulations and identical method of processing had previously been tested, that they comply with USP requirements for Class II plastics and that they are the same as being offered on this contract.
***Ten grams of plastic material-representative of each different plastic used in the fabrication of the components-shall be selected from each lot for the heavy metals test. Test shall be performed on the finished set.
****The unit of product for the leaching test shall be one pound of plastic sheeting. The sample size, as determined by the total number of pounds in the batch of plastic, and the specified inspection level shall designate the number of strips to be cut from the lot of sheeting for the leaching test.

4.2.4 Raw materials. A sufficient quantity of each lot of metal used in the construction of the needles shall be selected for chemical analysis.

4.2.5 Components. Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the sets to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated above. No test result shall fall below the minimum or exceed the maximum value specified. The sample unit for testing shall be the quantity necessary to perform all required tests one time each. Not less than two samples shall be taken regardless of lot size.

4.2.6 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

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4.3 Examination. The blood collecting and dispensing sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the following classification of defects. Examination shall include, but shall not be limited to, the classified possible defects listed in Table III.

Table III. Classification of Defects

Categories	Defects
Critical	
001	Missing, raised, or dislodged needle sheath.
002	Visually discernible leak in needle sheath or needle hub.
003	Missing, illegible, or incomplete markings on donor tubing.
004	Different numbers on same donor tubing.
005	Identical numbered donor tubing on two or more units.
006	Open or leaking bag port.
007	Missing or incorrect label on bag.
Major	
101	Needle not free of cracks, burrs, or silvers.
102	Needle outer surface not smoothly finished or free of pits, tool marks, scale or corrosion.
103	Needle lumen not free of obstructing material, scale,burrs, chips, dirt, deep pits, deep furrows, or die marks.
104	Needle point not keen, not straight, symmetrical, or properly formed, or not free of burrs, slivers, feathered edges, or other detrimental defects.
105	Donor tubing flattened with walls touching.
106	Bag outlet tubes collapsed (walls touching) so as to prevent entry into port.
107	Outlet port sleeves or plugs not removable.
108	Plastic material of bag or donor tubing not transparent or free of discoloration.
109	Surfaces of bag or donor tubing not smooth or free of bloom, brittleness, or tackiness.
110	Unit not free of foreign matter.
111	Side slits or holders missing, broken, or break on opening.
112	Hanger slot, holes, or strips missing.
113	Instructions for use missing, incomplete, or illegible.
114	Lot number missing, incorrect, incomplete, or illegible.

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Table III. Classification of Defects (Cont'd)

Minor	
201	Bag or tubing not free of gel marks, die marks, wrinkles, or particulate matter.
202	Donor tubing not free of minor kinks or distortion.
203	Donor tubing protruding into bag.
204	One or more digits of numbered tubing sealed into top of bag or under needle hub.
205	Bag label not free of wrinkles, creases, tears, or slits.
206	Printing defects or extraneous markings on back of label.

4.4 Tests. Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with this specification. In addition to performing the chemical analysis and physical tests, the supplier shall perform or have performed for him, at his expense, all bacteriological, biological, and hemotological tests, including in vivo tests, required by the applicable specifications.

4.4.1 Biological tests. The plastic bag and donor tubing shall be tested in accordance with the requirements of the U.S.P. for Class II plastic.

4.4.2 Heavy metals-extractables test. The plastic bag and donor tubing shall be cut such that no dimension exceeds 3 mm and eluted for either 72 hours at 50° C., 24 hours at 70° C., or 60 minutes at 121° C., in two menstrua: 0.9 percent saline and cottonseed oil. Each gram of the plastic material in each instance shall be covered with 5 ml. of the menstruum (recommended amount is 10 grams of plastic plus 50 ml. of solvent). The menstruum shall be collected, after agitation, at the end of the selected elution period and submitted to spectrographic analysis for barium, cadmium, and lead. The metals shall be reported in parts per million in the eluate. A value of 1.0 p.p.m. in the eluate shall be equivalent to extracting 1.0 milligrams (mg.) from 200 grams of test material in the elution period into 1000 ml. elution menstruum.

4.4.2.1 Alternate heavy metals extractable test. Accurately weigh a 5 - gram sample for each of the metals (barium, cadium or lead) into a 1 - oz. Vycor crucible. Reduce to ash in a muffle furnace at 650° C for eight hours. Let cool to room temperature and add 15 ml. of 1.1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50 - ml. volumetric flasks. Rinse the residue on the filter with deionized water and dilute to 50 ml. Prepare standard solutions of 1, 2, and 5 ppm for each of the metals, namely barium, cadmium, and lead. Determine the absorbances of these solutions at appropriate wave lengths using a 10X scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303, Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

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Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm barium, cadmium, or lead}$$

A = Absorbance
F = Factor = $\frac{\text{(Concentration standard in ppm Ba, Cd, or Pb)*}}{\text{absorbance}}$

W = Sample weight in grams
* The three standards are averaged to give ppm/unit A.

4.4.3 Vapor transmissivity test. The bag shall be filled with 450 ml. of isontonic salt solution and 63 ml. of anticoagulant solution, and shall be stored for 4 weeks at a temperature of 10° C., and an absolute humidity of approximately zero. At the end of this period, the contents of the bag shall be measured.

4.4.4 Leaching test. Cut 2-inch by 3-inch strips from the beginning, approximate middle, and end of each lot of plastic sheeting. Dry samples for 24 hours at 121° Fahrenheit (F.). Remove samples from oven and allow to cool in a dissicator over calcium chloride for 30 minutes. Accurately weigh samples to the nearest 0.1 mg. Suspend samples in 4-ounce jars containing 120 ml. of citrate phosphate dextrose solution (CPD Solution A-U.S.P.) and autoclave at 250° F. for 30 minutes. Remove samples, rinse fied water, and dry for 24 hours at 121°F. Remove samples and allow to cool in dissicator over calcium chloride for 30 minutes. Reweigh and calculate percent weight loss. As a standard, subject a strip of plastic from the same roll to all the above conditions, except do not suspend in citrate phosphate dextrose solution.

4.4.5 Bag suspension test.

4.4.5.1 Horizontal slot. With the bag suspended by the horizontal slot (by means of 1-1/4-inch-wide webbing having a weight of 0.81 oz. per linear yard), a 5-pound force shall be applied to bag. This force shall be maintained for 15 minutes. The bag shall then be examined for damage.

4.4.5.2 Circular holes and loops. With the bag suspended by means of one of the circular holes or loops at the top of the bag, a 5-pound force shall be applied to the bag. This force shall be maintained for 15 minutes. The bag shall then be examined for damage. This procedure shall be repeated for each of the circular holes and loops on the bag.

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4.4.6 Plastic bag leakage test. The plastic bag shall be filled with water to make a total volume of 600 ml. and squeezed between two parallel platens such that an internal pressure of 500 mm. of mercury at 25°C. is gradually developed and maintained for 10 minutes.

4.4.7 Fluid fill test. By means of its donor needle, the bag shall be filled by gravity from a reservoir, filled with a fluid at 37° ± 2° C., of viscosity comparable to that of blood at the same temperature (3.2 centistokes at 37° C.), and under a pressure of 135 mm. of mercury.

4.4.8 Pressure infusion test. To the bag, plus anticoagulant, shall be added 450 ml. of water cooled to 5°C. A recipient set (as specified in 3.2.1.1) without recipient needle, shall be connected to the bag. The filled bag (with attached recipient set) shall be squeezed between two parallel platens such that an internal pressure of 300 mm. of mercury is gradually developed with no leakage.

4.4.9 Test for security of attachment .

4.4.9.1 Donor tubing to bag. A force of 5 pounds shall be applied gradually to the juncture of the tubing with the bag and maintained for 10 minutes. This juncture shall then be examined for damage.

4.4.9.2 Donor tubing to needle hub. The tubing of the donor set shall be severed at the midpoint of the distance between the hub of the needle and the point of juncture with the bag. The portion of the donor set, so severed, containing the attached needle, shall be suspended with the tubing hanging vertically downward by securely clamping the cannula of the attached needle in a clamping device. A 1-pound weight, applied gradually, shall be suspended from the tubing for 10 minutes.

4.4.9.3 Needle cannula hub. First clamp hub in a vertical position so that the cannula extends upward. Lower a 3-pound weight gradually onto the point and hold for 1 minute. Then clamp hub in a vertical position so that the cannula extends downward. Gradually apply a 3-pound force onto the cannula and hold for 1 minute.

4.4.10 Sterility test. Samples of the finished blood collecting and dispensing sets shall be tested for sterility in accordance with the procedure of the U.S.P. for parenteral devices. The anticoagulant solution shall be tested for sterility in accordance with the sterility test method of the U.S.P. for liquids as required by FED-STD-142. The method of testing whereby the anticoagulant solution is drained from the bag through the donor tubing and needle, and the exterior of the needle is flushed with anticoagulant solution, satisfies the sterility test requirement for the set and the solution.

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4.4.11 Test for freedom from pyrogens. Samples of the finished blood collecting and dispensing sets shall be tested for freedom from pyrogens in accordance with the test method of the U.S.P. for transfusion and infusion assemblies. The anticoagulant solution shall be tested for freedom from pyrogens in accordance with the pyrogen test method of the U.S.P. as required by FED-STD-142. The method of testing whereby the anticoagulant solution is drained from the bag through the donor tubing and needle, and the exterior of the needle is flushed with anticoagulant solution, satisfies the pyrogen test requirement for the set and the solution.

4.4.12 Safety (toxicity) test. Samples of the finished blood collecting and dispensing sets shall be tested for safety in accordance with the safety test method of the U.S.P. for transfusion and infusion assemblies.

4.4.13 In-vivo survival test. See Appendix A. Sample units made from each lot of plastic sheeting shall be subjected to red cell survival study. Three tests, as described in Appendix A, shall be performed. If red cells survival is not less than 70 percent in each test after 21 days of continuous storage at 5° ± 1° C., the material shall be acceptable. If one determination is less than 70 percent, a fourth test shall be performed. If this is satisfactory, the material is acceptable.

NOTE: Among a population of normal subjects are some individuals who destroy their own stored red cells at an accelerated rate. No accepted known means is available for detection and exclusion of these donors other than the red cell survival study. (Dern, R.J. et. al. Studies on the preservation of human blood. I. Variability in erythrocyte storage characteristics among health donors. J. Lab. Clin. Med. 67: 95 5-65, 1966).

If there is a reason to believe that two such donors have been used to test a lot of plastic, six more in vivo survival studies may be performed. Any study falling more than 2 S.D. below the mean of the group of 10 is excluded. If the average of the remaining studies is 70 percent or more, the material is acceptable. Otherwise, the lot is rejected. Under the direction of a physician, the in vivo survival tests are carried out by techniques listed in Appendix A.

4.4.14 Label adhesion test. When adhesives are used to secure the label to the bag, the following test shall be performed. Not less than five labeled bags shall be allowed to dry at room temperature for a period of not less than 5 days. The labeled bags shall be completely submerged in tap water maintained at a temperature of 75° Fahrenheit (F.), (plus or minus 5° F.) for a period of not less than 48 hours.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One can containing six sets, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	8 units	16 units

5.1.2.1 Packaging variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit preparation. Each set and components shall be neatly arranged and folded (when required). Tubing shall be coiled into a coil of such diameter as to prevent kinking or acquiring a permanent set.

5.1.3.2 Unit packaging.

5.1.3.2.1 Inner container. Six sets shall be packaged in a bag fabricated from polyethylene and having an average wall thickness of not less than 0.002 inch, with no wall thickness at any point below 0.00125 inch. The sets shall be packaged within the bag so as to prevent kinking of the tubing and to retain sterility. Closure shall be effected by twisting the opening of the bag and securing in position within the outer container.

5.1.3.2.2 Outer container. Each inner container, 1 unit, shall be further packaged in a can conforming to type I of PPP-C-96 and fabricated of 1.25-pound hot dipped tinplate or 0.25-pound electrolytic tinplate. Interior coating shall conform to paragraph 3.2.1 thereof. The can shall be provided with an aluminum pull-top lid.

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5.1.3.2.3 Completed package. The complete unit package (inner and outer containers) shall afford the degree of protection as specified herein.

(a) Size. The completed package shall be of such a size as to utilize the minimum of storage or shipping space.

(b) Treatment. The materials used for packaging shall be of such a nature or treated to prevent growth of mold or spore bearing organisms. Chemical fungicides will not be permitted unless evidence can be submitted that they do not penetrate the bag.

(c) Shelf life. The sets being supplied shall be fabricated of such material and so packaged as to maintain units serviceable for a period of not less than four years when subjected to military depot storage conditions at temperatures below 86°F.

(d) Sealing. The completed package shall be hermetically sealed and shall be tamperproof and prevent opening or reclosure without destroying the seal.

5.1.3.3 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic, and fabricated from not less than 200 pound test fiberboard. Closure shall be adequate to prevent spilling of contents under normal handling.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be a box of appropriate size constructed in accordance with PPP-B-636, class weather-resistant, compliance symbol V3c or V3s, style RSC.

5.2.1.1.1 Closure. Closure shall be as specified in the appendix of PPP-B-636 and water-proofing shall conform with 30.4.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic and fabricated from not less than 275 pound test fiberboard. Method II closure specified in PPP-B-636 shall be utilized.

5.3 Marking.

5.3.1 Unit package. Each can shall be lithographed with the information as specified for the unit package in MIL-STD-129. In addition, marking shall include the lot (control) number, contract number, name of contractor, requirements of statute, instructions for use (see 3.8), type II shelf life markings as specified in MIL-STD-129, and the method of sterilization (G-Gas, R-Radiation, S-Steam, C-Cold, D-Dry heat sterilization) shall be shown following the level of packaging and date of packaging: Example: "A-2/73-G". Packaging shall be on

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5.3.1 Unit package (con't).

or after indicated sterilization date. When the contractor is not the manufacturer, the name of the manufacturer shall also be shown. The lot (control) number, type II shelf-life markings, and contract number may be applied to the bottom lid of the can. In addition, marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

5.3.2 Intermediate package. Intermediate package shall be marked in accordance with MIL-STD-129. Lot (control number contract or purchase order number and name of contractor shall be shown. In addition, type II shelf life marking, the method of sterilization and the legend as specified in 5.3.1 shall also be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number, type II shelf life marking, the method of sterilization and the legend as specified in 5.3.1 shall also be shown.

6. NOTES

6.1 Intended use. The blood collecting and dispensing bag is intended for collecting, storing, processing and administrating 450 ml of blood or centrifuged or resuspended blood cells with minimum hazard of bacterial contamination.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable level of packing. (see 5.1 and 5.2).

6.3 Federal Stock Number (FSN) coverage. This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.4 This specification covers the following item listed in the Federal Supply Catalog:

Federal Stock Number	Item Identification
6515-079-9483	BLOOD COLLECTING-DISPENSING BAG AND DONOR SET, With 63 cc Anticoagulant, Disposable, 6s.

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Custodians:
Army - MD
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Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1692

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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APPENDIX A

POST-TRANSFUSION SURVIVAL STUDIES OF STORED RED CELLS TAGGED WITH CR^{51}

10 Scope

This appendix details the test procedure for the determination of the post-transfusion viability of human erythrocytes by the radioactive chromium method.

10.1 Sodium Chormate Tagging

One-half ml. of sterile sodium chromate solution, as obtained from the Abbott Laboratories, Oak Ridge, Tennessee, containing approximately 500 microcuries of Cr^{51} as $Na2Cr^{51}04$, is diluted to a volume of 5 ml with 0.9% sodium chloride solution.

Twenty-eight ml. of citrated stored blood are removed and mixed with a volume of the dilute sodium chromate containing approximately 60 microcuries. After gentle mixing, the blood is permitted to incubate in a 37° C. water bath for 20 minutes.

NOTE: It has been found that a maximal amount of Cr^{51} is taken up by the erythrocytes within this period of time, but the uptake varies between 85 and 95% of the amount available, depending on the hematocrit of the blood, i.e., the higher the hematocrit the larger the volume of red cells and the higher the Cr^{51} uptake, provided the volume of the sample of whole citrated blood remains constant.

The specific activity of the sodium chromate must be sufficiently high so that the resulting concentration of chromium will not exceed 3 micrograms per ml. of packed red cells.

10.2 Infusion of Chromated Blood

At the end of the period of incubation, the blood is gently but thoroughly mixed, and 10 ml. are withdrawn into a calibrated syringe and immediately and rapidly injected into the recipient. At the same time, a portion of the remaining chromated blood is centrifuged and the plasma is separated.

To obtain sharp and reproducible results, the time of injection of the 10 ml. of tagged red cells should be as short as possible, between 7 and 10 seconds. Under these conditions, and when a normal subject is used as a recipient, the mixing time of the transfused red cells is 3 minutes or less.

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10.3 Measurement of Radioactivity

The uptake of Cr^{51} by the red cells is determined by the differential count on diluted samples of whole blood and plasma, as follows: a 1:100 dilution of the remaining chromated blood in distilled water and a 1:25 dilution of the separated plasma in saline solution are made.

Radioactivity measurements are carried out by counting gamma emission from a 3 ml. sample of the diluted material placed in a flat bottom test tube, using a well-type scintillation counter provided with a thallium activated sodium iodide crystal. The radioactivity is expressed as counts per minute per ml. (c pm/ml.) of packed cells, after correction for dilution and subtraction of plasma activity.* To determine the survival rate of transfused cells, samples of 3 ml. of undiluted blood are used for radioactivity measurements; these samples of blood are centrifuged, after being placed in the flat bottom tubes, prior to counting.**

* All values are corrected for decay of radioactivity when indicated.

** One ml. of the plasma is diluted with 2 ml. of distilled water for purpose of counting and correction.

10.4 Determination of Erythrocyte Survival

The survival of transfused cells is determined by the use of the following formulae:

$$\text{100\% survival of transfused red cells} = \frac{\text{Total cpm of transfused red cells}}{\text{Total red cell vol. ml.}}$$

$$\text{Apparent \% survival at any time} = \text{cpm/ml. of whole blood} \times \frac{100}{\text{hematocrit}} \times 100$$

$$\frac{}{\text{100\% survival}}$$

The blood volume determination can be carried out either by measuring directly the volume of red cells with the P^{32} method, or by the measure of the plasma volume with the Evans blue technique and the determination of the hematocrit, multiplied by the factor of 0.91 to obtain the total body hematocrit.

Determinations are carried out at 20 minutes after completion of the transfusion and at 24 hours after transfusion. Thereafter, the counts are carried out at intervals of 1 or 2 days for 10 days.

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10.5 Computations (based on counts per minute per milliliter)

REMAINING CHROMATED BLOOD:

$\frac{\text{cpm of diluted blod x 100}}{3}$ = cpm/ml. of blood

$\frac{\text{cpm of diluted plasma x 25}}{3}$ = cpm/ml. of plasma

$\text{cpm/ml. of plasma x } \frac{(100 - \text{Hct})}{100}$ = cpm/ in plasma/ml. of blood

cpm/ml. blood - cpm/in plasma/ml. blood = cpm/on cells/ml. of blood

cpm/in cells/ml. blood x 10 = ACTIVITY INJECTED ON THE CELLS

$\frac{\text{cpm in cells/ml. of blood x 100}}{\text{cpm/ml. of blood}}$ = % Uptake by the red cells

POST INJECTION SAMPLES:

$\text{cpm/ml. plasma x } \frac{(100 - \text{Hct})}{100}$ = cpm in plasma/ml. of blood

$\frac{\text{cpm in blood}}{3}$ - cpm in plasma/ml. blood = cpm in cells/ml. of blood

$\frac{\text{cpm in cells/ml. of blood}}{\frac{\text{Hct}}{100}}$ = cpm/ml. of red cells in sample

10.6 References

Max M. Strumia, M.D., Lawrence Taylor, Albert B. Sample, Louise S. Colwell, Ann Dugen: Uses and Limitations of Survival Studies of Erythrocytes Tagged with Cr51, Blood, Vol. X No. 5, 429 May 1955.

John G. Gibson II, M.D.: A Method Employing Radioactive Chromium for Assaying the Viability of Human Erythrocytes Returned to the Circulation after Refrigerated Storage, J. Lab & Chin. Med., 46:679 (1955).

Note:

The hematocrit used in the determination referred to above is the one employing a capillary tube with very high centrifugal force, resulting in near absolute packing of red cells. Hence, no correction for trapped plasma need to be made (see: "An improved Micro Hematocrit Method", by Max M. Strumia, et al, The American Journal of Clinical Pathology, Vol. 24, No. 9, September, 1954).

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filter shall be re-examined with a light upon completion of this test, the

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